

## **Rohini Kolli**

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### **PROFESSIONAL SUMMARY**

Data Professional with 5+ years of experience in data engineering, analytics, and applied machine learning across manufacturing and financial domains. Skilled in building scalable ETL pipelines, orchestrating workflows, and deploying production-grade ML systems in cloud environments. Proficient in Python, SQL, Spark, Hadoop, and orchestration tools (Airflow, Metaflow) with hands-on experience in AWS (SageMaker, EMR, Glue) and Azure (Data Factory, Synapse, Databricks). Adept at implementing MLOps best practices, managing feature stores, executing batch inference, and building end-to-end model pipelines. Strong foundation in statistical and ML techniques including logistic regression, Naive Bayes, SVM, decision trees, neural networks, and NLP to deliver actionable insights and measurable business impact.

### **SKILLS**

**Programming & ML Libraries:** Python (Pandas, NumPy, scikit-learn, TensorFlow, PyTorch), SQL, R, Jupyter Notebook, Git

**Machine Learning & AI:** Logistic regression, Naive Bayes, SVM, decision trees, neural networks, NLP, anomaly detection, A/B testing, statistical modeling

**Data Engineering & Big Data:** Hadoop, Spark, PySpark, Presto, Hive, Snowflake, PostgreSQL, Azure Data Factory, Databricks, AWS Glue, AWS EMR, DBT

**Cloud & MLOps:** AWS SageMaker, Azure ML, CI/CD, Docker, Jenkins, DevOps, feature store, batch inference, model pipeline orchestration

**Visualization:** Power BI, Tableau, Plotly, ggplot

### **PROFESSIONAL EXPERIENCE**

**Teaching Assistant – Data Visualization & BI Tools | Northeastern University | Sep 2023 – Dec 2023**

- Led interactive lab sessions on building KPI-driven dashboards in Power BI and Tableau, emphasizing drill-through filters, dynamic visuals, and actionable insights.
- Mentored graduate students on SQL queries, Python-based data transformations, and dashboard performance optimization to enhance analytical outcomes.

**Manufacturing Data Professional (Co-op) | Veeco Instruments | Jan 2023 – Jul 2023**

- Built scalable ETL pipelines in Azure Data Factory and Databricks to ingest high-frequency FabGuard sensor and metrology data, implementing data quality checks, anomaly detection, and schema mapping for predictive analytics and process monitoring.
- Designed PySpark/Hadoop transformations to clean and standardize IoT telemetry streams from 10+ fabrication tools, enabling SPC rule evaluation, predictive maintenance, and batch inference workflows.
- Developed Power BI dashboards connected to Azure Synapse, integrating outputs from ML models such as logistic regression, decision trees, and XGBoost for visualizing SPC metrics, chamber utilization, yield variation, and process drift. Enabled engineers to monitor both statistical trends and model-based risk scores in real time, reducing time-to-insight by 40%.
- Orchestrated data workflows with Airflow alongside Azure Data Factory and deployed ML models in Azure ML with CI/CD pipelines in Jenkins, incorporating feature store integration and batch inference scheduling; workflows transferable to AWS SageMaker.

**Data Scientist | HCL Technologies | Feb 2019 – Jan 2022**

- Designed and maintained large-scale ETL pipelines in Python, SQL, and PySpark to integrate transactional, behavioral, and campaign datasets into a unified banking data model.
- Migrated legacy SSIS workflows to Azure Data Factory and Snowflake, developing parameterized, reusable pipelines with dynamic file handling, improving data availability by 40%.
- Built fraud detection and churn prediction models using scikit-learn (logistic regression, Naive Bayes, SVM) and TensorFlow, improving fraud detection accuracy by 18% and reducing false positives by 15%.
- Orchestrated ML and data workflows using Metaflow and Airflow, integrating with AWS Glue-compatible ETL processes, feature stores, and batch inference pipelines.
- Created DBT models in Git for centralized KPIs and developed interactive Tableau dashboards with cohort analysis and KPI tracking, enabling faster, data-driven decision-making.

#### **Data Analyst | Horizon Informatic Services | Jun 2016 – Feb 2019**

- Built SQL Server and Python-based ETL pipelines to aggregate customer, loan, and repayment datasets, improving completeness by 30% and reducing manual integration time.
- Automated regulatory reporting via SQL Server Agent and Excel VBA, reducing recurring reporting hours by 25+ per month.
- Developed Tableau dashboards for credit risk, churn, and fraud analysis, incorporating anomaly detection logic to improve fraud capture by 25%.

#### **ACADEMIC PROJECTS**

1. **Real-Time Transaction Fraud Detection Pipeline:** Built an end-to-end ML pipeline using PySpark and scikit-learn to detect fraudulent credit card transactions in real time. Ingested streaming data via Kafka, processed features in Spark, and deployed models to AWS SageMaker with batch inference scheduling, achieving 98% recall on fraud cases.
2. **Hospital Readmission Prediction:** Designed a predictive analytics workflow to forecast patient readmissions using logistic regression, random forests, and neural networks in TensorFlow. Integrated feature engineering scripts in Airflow, enabling automated retraining every quarter.
3. **Customer Churn Prediction Platform:** Designed and implemented an automated churn prediction pipeline using Azure Data Factory, Databricks, and scikit-learn. Ingested transactional and behavioral data, engineered features for model training, and deployed the model for monthly batch inference with results visualized in Power BI.